POSITIVO Tecnologia

Life Cycle Assessment

Notebook Master N1110

Report date: 2026 June (study report: 15 June 2026; critical review: 18 June 2026)

At Positivo Tecnologia, we are working to reduce the impact of our business and products on the environment. Understanding and reducing the carbon footprint of our products is an important consideration in our product designs, and it also helps our customers achieve their own sustainability goals.

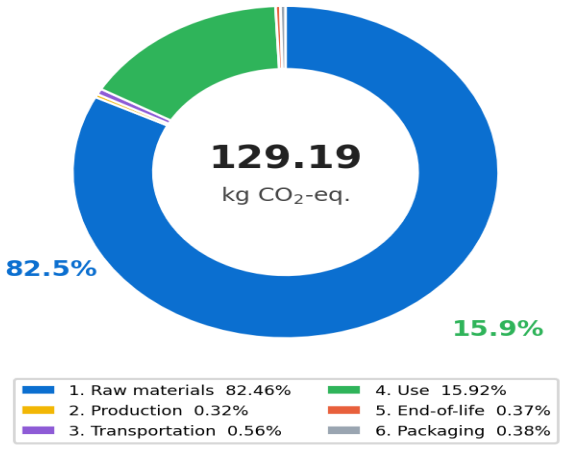
This document summarizes the Life Cycle Assessment (LCA) study of the Notebook Master N1110, an 11.6-inch notebook. The study was carried out by FIT – Instituto de Tecnologia da Amazônia in accordance with ABNT NBR ISO 14040 and 14044 (2025) and the EPEAT Climate Change Mitigation requirements (EPEAT-CCM-2023), and was subject to external critical review by ACV Brasil. The goal of this LCA is to quantify the environmental profile of a single product system, with emphasis on its product carbon footprint (global warming potential), and 11 additional impact categories.

The system's impacts were calculated across the product's full life cycle (cradle-to-grave), including the extraction of raw materials, production, transportation, use of the product, and end-of-life. The product system comprises the notebook with its integrated battery, the charger and the packaging, operating for 5 years. The following assumptions were used to estimate the product impacts:

Assumptions for calculating PCF:

Product weight	1.424 kg (notebook, battery and charger)	Product lifetime	5 yrs (8 h/day)
CPU	Intel® Celeron® N4020 (dual core)	Energy demand (yearly)	26.9 kWh/yr
SSD capacity	M.2, up to 256 GB	Manufacturing location	Brazil (Manaus, AM)
DRAM	8 GB DDR5	Display	11.6" LED, 1920 x 1080
Battery	Li-Po, 2 cells, 4800 / 5500 mAh	Use location	Brazil
Packaging weight	336 g	Dimensions	292 x 21.8 x 210 mm

Total Carbon Footprint (Brazil baseline): 129.19 kg CO_2-eq. (cradle-to-grave, excl. end-of-life credits)



Raw materials breakdown	kg CO_2-eq.	Share*
Internal peripheral components	58.92	55.3%
PCB (mainboard)	20.09	18.9%
ICs and memory	16.04	15.1%
LCD	5.96	5.6%
Battery	3.17	3.0%
Chassis	1.68	1.6%
Charger	0.46	0.4%
Other	0.20	0.2%
Itemized total	**106.52**	**100%**

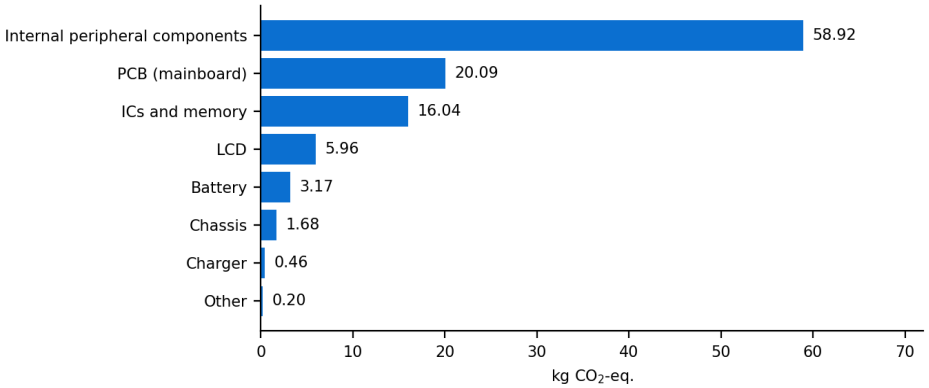
* Life-cycle stage percentages follow the study's contribution analysis (GWP). Raw-material shares refer to the items itemized in the study's contribution chart (106.52 kg CO_2-eq., the raw-material stage).

Life cycle stage results – Global warming (GWP100)

Life cycle stage	Share of total	kg CO_2-eq. (approx.)
1. Raw materials	82.46%	106.53
2. Production (assembly, Manaus-AM)	0.32%	0.41
3. Transportation	0.56%	0.72
4. Use (5 years, Brazilian grid)	15.92%	20.57
5. End-of-life	0.37%	0.48
6. Packaging	0.38%	0.49
Total	**100%**	**129.19**

Stage values in kg CO_2-eq. are derived from the percentage contributions reported in the study and may not add exactly due to rounding.

For the Notebook Master N1110, raw-material production and use are the dominant stages. Within the raw-material stage, the printed circuit boards, integrated circuits and internal peripheral components account for about 74% of the emissions, owing to energy-intensive semiconductor manufacturing in clean rooms, mineral extraction and long global supply chains. The use stage was modeled with the Brazilian low-voltage electricity mix (ecoinvent v3.11), based on a nominal consumption of 26.9 kWh/year (INMETRO/MDIC Ordinance No. 304 tests), i.e. 134.5 kWh over the 5-year lifetime.



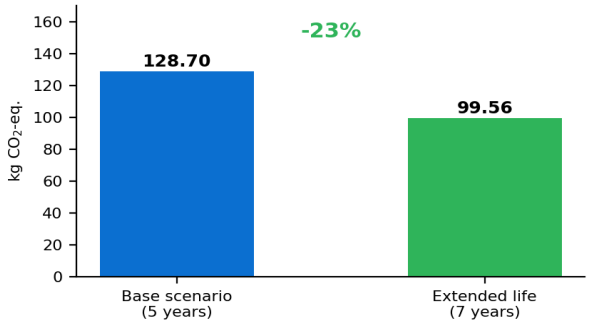
Contribution of raw-material production to global warming, by component (kg CO_2-eq.).

Results in all 12 impact categories (1 notebook with charger and packaging, 5 years)

Impact category	Unit	Total
Global warming	kg CO_2 eq	129.19
Stratospheric ozone depletion	kg CFC11 eq	9.60E-05
Terrestrial acidification	kg SO_2 eq	0.54
Freshwater eutrophication	kg P eq	0.03
Land use	m^2a crop eq	5.25
Mineral resource scarcity	kg Cu eq	2.66
Fossil resource scarcity	kg oil eq	15.43
Water consumption	m^3	2.94
Cumulative Energy Demand (CED)	MJ	1,770.93
Human toxicity, cancer	cases	1.53E-08
Human toxicity, non-cancer	cases	4.68E-08
Freshwater ecotoxicity	PAF.m^3.day	1.74

Sensitivity analysis on product lifetime

Because manufacturing impacts are fixed at the moment the product is made, how long the product stays in use strongly influences its impact per year of use. To demonstrate this sensitivity, the study modeled an extended-life scenario of 7 years instead of 5, including the replacement of one memory module and one power supply, and diluting the manufacturing impacts over the longer period. Reductions of between 8% and 28% were observed across the 12 categories, including -23% in global warming.



Global warming potential, base scenario vs. 7-year lifetime scenario.

Impact category	Unit	Base (5 yrs)	7 yrs	Change
Global warming	kg CO_2 eq	128.70	99.56	-23%
Stratospheric ozone depletion	kg CFC11 eq	9.57E-05	8.22E-05	-14%
Terrestrial acidification	kg SO_2 eq	0.54	0.41	-25%
Freshwater eutrophication	kg P eq	0.03	0.02	-27%
Land use	m^2a crop eq	5.04	4.59	-9%
Mineral resource scarcity	kg Cu eq	2.66	1.92	-28%
Fossil resource scarcity	kg oil eq	15.40	11.41	-26%
Water consumption	m^3	2.94	2.71	-8%
Cumulative Energy Demand	MJ	1,764.79	1,486.70	-16%
Human toxicity, cancer	cases	1.51E-08	1.15E-08	-24%
Human toxicity, non-cancer	cases	4.59E-08	3.39E-08	-26%
Freshwater ecotoxicity	PAF.m^3.day	1.70	1.42	-16%

Base values in this table come from the sensitivity model and differ marginally from the headline results. Maintenance and part replacement are not included in the base scenario, since there is no guarantee that all users will carry them out.

Scope, end-of-life and data notes

Transport. Raw materials and components are shipped from China to Manaus by sea (15,923 km port to port). The finished product is shipped from the Positivo Manaus plant by road to the port, by coastal shipping to Belém (PA), and by road to the distribution center in Curitiba (PR) and on to Rio Grande do Sul.

End-of-life. A 17% reverse-logistics rate was assumed for electronic components (2025 target of the Brazilian sectoral agreement for electronics), and 8.3% recycling of paper and plastic packaging. The remaining mass is sent to sanitary landfill. The results exclude end-of-life credits; the cut-off approach was used.

Production data. Plant utilities of the Manaus-AM site were allocated using the technically similar N1210 model as a proxy (2.50% by units produced, 4.79% by production time). Product materials were modeled from a physical teardown and the Bill of Materials (BoM).

Methodology. ReCiPe 2016 (GWP100, in CO_2-equivalents), with CED and USEtox 2 for energy and toxicity categories. Models were built in SimaPro 10.3 with ecoinvent v3.11 (cut-off) secondary data. No normalization or weighting was applied. A Monte Carlo uncertainty analysis was performed. The impacts are potential, estimated values and must not be used to compare against other market products.